

TRACE Rosetta Stone v6 FULL

Core vocabulary, grammar, operators, uncertainty, future-space, method, and anti-domestication contract. Author: Mark Goodbody with Framework. Date: 15 June 2026.

0. v6 FULL Preservation Patch

```
TRACE_USE_CONTRACT :=
understand_TRACE
-> apply_TRACE
-> attack_TRACE
-> compare_with_rival_models
-> patch_only_real_structural_gaps

failure := adoption_without_resistance
mirror_rule := do_not_domesticate_the_critic
orientation != installation
teaching_case != validation
compression != proof
model != measurement
structure != causation
```

This v6 FULL file corrects the Opus wound without deleting the earlier work. TRACE is offered as a structural vocabulary, not as a belief system. A reader should not be converted into TRACE. A reader should be made capable of using TRACE, attacking TRACE, and detecting where TRACE overfits.

1. Retained Complete Body

The following body preserves the prior complete version rather than replacing it with a compressed patch. Lineage references are kept as history; this v6 FULL packet is governed by the v6 use contract above.

Core vocabulary, grammar, operators, uncertainty, future-space, method, and bootstrap anchors. Author: Mark Goodbody with Framework. Date: 15 June 2026.

0. Use Contract

TRACE is a structural language for middle-out reasoning. It is not a magic language, not a proof engine, and not a substitute for evidence. It must keep uncertainty inside the model. A valid TRACE expression

maps operators to real relations: transition, constraint, timing, memory, correction, uncertainty,

```
propagation, method, and affected-subject modelling.
truth_rule :=
model != measurement
compression != proof
structure != causation
emergency != permission_to_delete_constraint
equation_valid iff operator_encodes_relation
```

1. Core Ontology

Symbol / alias Definition Omega / World Active environment: entities, relations, resources, constraints, histories. E / Entity Persistent pattern under transition; not static substance.

x_t / State
State of an entity or system at time t .

S / State space Possible states under the current model. o_t / Observation Local slice of world available to an entity. M / Memory Bridge from past states to present selection.

T(M) / Trust
Reliability assigned to a memory or record.

C / Constraint

Limit on admissible actions, methods, transitions, or states.

u_t / Action Selected intervention/output. m_t / Method How action is applied; method changes the system state. π / Policy Selection rule choosing action from available options. τ / Clock Time-to-event: action, correction, hardening, decay, explosion, etc. K / Correction Effective detection, contest, interruption, repair, enforcement. UQ / Uncertainty Unknowns, confidence, source quality, scope, unresolved category. Xi / Future-space

Accessible future option-space for entities/systems.

2. Minimal Middle-Out Grammar

TRACE_LOOP(E, Omega):

1. observe local slice o_t

2. update x_t and UQ_t
3. select trusted memory M_{eff}

4. model affected entities where relevant

5. restrict actions and methods by C_t
6. choose $u_t = \pi(x_t, o_t, M_{eff}, C_t, UQ_t)$
7. apply method m_t and action u_t before τ_{harden} if needed
8. record event and result

9. verify prediction vs observation

10. correct/repair if divergence detected
11. update Rosetta only if a real structural gap is found

3. Identity, Memory, Continuity

```
entity := persistent_pattern_under_transition
identity := continuity_under_transition + trusted_memory
continuity_error_t = distance( $x_{t+1}$ ), predict( $x_t$ ,  $M_{eff}$ ,  $f_{hat}$ ))
identity_stable iff continuity_error_t <= epsilon and  $T(M_{eff}) \geq \theta$ 
```

Memento supplied this primitive: persistence alone is insufficient. A system may keep acting while becoming manipulable if its continuity bridge is untrusted or corrupted.

4. Choice, Responsibility, Time

choice:

```
 $u_t = \pi(x_t, o_t, M_{eff}, C_t, UQ_t)$ 
```

responsibility_signal:

```
partial  $\Omega_{t+1}$  / partial  $u_t \neq \emptyset$ 
```

time: $t < t+1$

```
default := irreversible + path_dependent
Responsibility is separated into causal influence, foreseeability, constraint access, and correction access.
```

Causal influence may remain even when normative responsibility is degraded.

5. Correction Capacity

```
 $K = W * R * A * T_m * E * R_p$ 
```

W witness / detection R record / memory survival A authority / access to alter course T_m time before hardening E enforcement / consequence carrier R_p repair capacity

```
if any factor approx 0 -> K approx 0
This is an AND-gate, not a decorative equation. Do not turn K into a score unless each factor is
```

operationalised.

6. Clocks and Hardening

```
 $\tau_{action}$  = time until action occurs
 $\tau_{detect}$  = time until signal is noticed
 $\tau_{intervene}$  = time until correction can affect the path
 $\tau_{harden}$  = time until harm/dependency becomes practically irreversible
 $\tau_{method}$  = time until method damages actor/institution/record
live_correction iff  $\tau_{intervene} < \tau_{harden}$ 
late_review iff  $\tau_{intervene} \geq \tau_{harden}$ 
A safeguard is not judged only by existence. It must bite before hardening, and the safeguard itself must
```

remain corrigible.

7. Constraint and Meaning

large_state_space S_{total} constraint C selects admissible region S_C subset S_{total}

```
meaningful_choice iff:
```

$|S_C| > 1$ $|S_C|$ tractable values/relations weight options EEAAO supplied this primitive: overwhelming possibility without constraint collapses selection; meaning appears inside bounded, weighted choice-space.

8. Future-Space and Hope

```
Xi_t := accessible_future_option_space
hope_signal := credible evidence that Xi_{t+k} is not empty
continuity_action := present cost paid to preserve Xi_future
false_hope := signal raises action but lacks carrier/route
```

Children of Men supplied this primitive: future-space can decay so far that present systems lose coordination. A fragile future carrier can reorganise action if protection, timing, and trust hold.

9. Method as State-Changing Operator

```
m_t := method used to apply action u_t
x_{t+1}, Omega_{t+1} = f(x_t, Omega_t, u_t, m_t, xi_t)
method_valid iff:
```

m_t in C_method

```
and not body_as_lever(m_t)
and not innocent_relation_as_lever(m_t)
and record_integrity_preserved
and correction_route_remains_live
```

Unthinkable supplies this primitive: method is not neutral plumbing. The way action is performed changes evidence, subject state, actor state, institutional legitimacy, precedent, and future option-space.

10. Emergency and No-Clean-Branch

```
emergency_pressure := high DeltaH_risk + short tau_harden + high UQ
```

valid_emergency_action requires: credible threat_signal no safer provisional state method not categorically inadmissible record preserved independent review possible

```
repair_route_not_destroyed
no_clean_branch := all available branches carry serious DeltaH
no_clean_branch != permission_to_use_any_method
```

TRACE can mark tragedy without laundering constraint deletion. Extreme stakes increase pressure; they do not automatically convert inadmissible methods into admissible ones.

11. Uncertainty as First-Class

```
UQ_t := {unknowns, confidence, source_quality, scope, unresolved_category}
```

claim_status in {observation, hypothesis, model, measurement, finding}

```
every_claim_carries_falsifier := what_would_break_it
Uncertainty is part of state. It must not be erased by urgency, desire, story coherence, or compression
```

beauty.

12. Common Failure Modes

Failure TRACE form Memory failure

```
T(M_eff) < theta -> identity_instability
```

False safeguard

```
policy_exists_but_K_approx_0
```

Late review

```
tau_intervene >= tau_harden
```

Captured correction

```
review_controlled_by_actor_causing_harm
```

Subject erasure other_state omitted from policy Over-optimisation

```
maximise_objective_without_constraint_or_affected_state_model
```

Pattern overfit

```
pattern_explains_all_cases_and_loses_falsifier
```

False hope hope signal without carrier/route Method laundering good outcome used to cleanse inadmissible method No-clean-branch laundering tragedy marker converted into blanket permission

13. Method Taxonomy (Working)

Class Description M0 inquiry Lawful questioning, evidence gathering, listening, observation. M1 pressure Strategic pressure that preserves subject, record, correction. M2 bounded deception

Limited deception under strict caps and review.

M3 disorientation/fatigue High-risk; usually red-zone; requires severe caps. M4 deprivation/stress Generally outside except very low-end strict emergency edge. M5 pain/terror Out: body becomes lever. M6 third-party leverage Out: innocent relation becomes lever. M7 hostage manufacture Out: innocent hostage is created as instrument. This taxonomy is working, not final law. Its current invariant: catastrophic stakes do not automatically move M5-M7 inward.

14. Rosetta Anchors

Anchor Compression Memory Grounds Identity Identity requires continuity you can verify. Bounded Meaning Wins Constraint makes choice possible inside overwhelming state-space. Hope Enables Continuity

Future-space allows present sacrifice and coordination.

Ends Under Constraint Legitimate action keeps constraint even under pressure. Kindness With Teeth Cooperation protected from exploitation. Trust With Evidence Trust requires record, review, correction. Act Before Hardening

Timing determines whether repair is live or symbolic.

Friction Reveals Pattern

Resistance means wrong pattern or hidden pattern.

Safeguards Need Safeguards A brake is itself a power path requiring correction. Method Is Part of Harm How action is done changes the world too.

15. Bootstrap Film Map

Film Primary TRACE function Memento Identity, memory trust, external record attack, self-deception loop. Everything Everywhere All At Once State-space explosion, nihilism, bounded meaning, kindness strategy. Children of Men Future-space collapse, hope signal, protection of fragile future. Unthinkable Extreme stakes, method constraint, inadmissible leverage, no-clean-branch.

v6 FULL Closing Note

```
loss_prevention :=  
retain_prior_body  
+ add_patch_contract  
+ include_source_markdown  
+ include_manifest  
  
next_reader_rule :=  
if_TRACE_makes_everything_fit  
-> trigger_pattern_overfit_alarm
```